

ActInf Textbook Group ~ Cohort 2 ~ Meeting 18

(Chapter 8, part 1)

Source: [YouTube](#) Playlist: Uncategorized Duration: 53:31 | Uploaded:

Unknown Transcript method: auto_caption

hello it's March 1st 2023 we're in cohort 2 of the paradol textbook group we're in meeting 18. our first discussion on chapter eight first Michael if you wanted to just raise the contacts or question that you wanted to begin with and we'll just pick up there you're I'm not sure if you're asking sorry my microphone was off yes yeah so uh the map territory fallacy I was wondering what this is and whether you could perhaps elaborate a little bit on that fantasy all right I'm moving it to chapter nine even though we're in chapter eight because we're gonna bring in uh figure from chapter 9. um oh you want to give a first thought on on this yeah sure um actually map territory fallacy is um one of the fallacies uh that is widely used to sum up somehow criticize a free energy principle uh because uh as uh the concept made the name of the concept may suggest some believe that it's a kind of fallacy to somehow conflate the map with the territory so by this I mean that free energy principle claims that uh well we have so very informally we have a map of the principle or the theory or whatever and the territory that it applies to so the system or the agent that we're trying to model but uh on the other hand by claiming that the system or the agent actually does the kinds of inference that recovers the map itself uh some people believe that this kind of conflation that map converts into territory or the U vice versa is a fallacious claim so we need to separate map from the territory but uh in this uh recent paper on the map territory fallacy fallacy ramstead at all uh argued that actually this is not a fallacy at all it is the structure of these uh of the fep theory so in other words we're well aware that there is a kind of uh inference as if epistemology happening here so uh yeah but it's a bit more extensive they've lined up some number of arguments to defeat this map territory fallacy fallacy and why it's not fallacious at all so yeah that's basically the gist of it in my opinion does it have to do with a distinction between model and process and the model is the maybe the model is the map and the no the other way the process is the map and the model and the external world I understand now the standard world is the is the what is it the process is the external world and the model is the um I mean the generative model of the

agent itself right yeah it definitely relates to that but on the other hand it also relates to this concept of self-evancing which is a way to describe the mechanism by which uh the model in the environment and or let's say the agent in the environment models itself according to the generative models so it's a kind of uh as the name suggests a self-referential way of modeling the system in the environment yes it's definitely a roller coaster and not a resolved one but still we can look to some of these key recent works so the map territory fallacy would be just like it's been since the classical borgas writing this realization like the map is not the territory kind of related like all models are useful or some models are useful but not are true like correlation is not causation this sort of broad space of aphorisms that's basically like instrumentalism isn't realism as we might say it so that's the map territory fallacy would be making the fallacy that the map is the territory I made a linear regression between Cancer and heart disease Ergo they are actually causal that way and when we're using octave inference framework one might be slightly more tempted than the linear regression to say well because I've generated a model that has a structure that seems to articulate certain Natural Things this is like an explanation it might even have predictive power and so that might lead somebody to make this kind of a fallacy that the map is the territory then ramstead at all have kind of agreed and Amplified which is saying actually when when you use the map to criticize the kind of cognitive modeling in the fep and Bayesian mechanics you're acting fallaciously and there's several angles on that unpack more in the paper one angle is you might be um chopping off more than you want to like you may realize it or not your critique May extend to Simply all quantitative modeling and many people want to like exercise fep from the space of scientific modeling in good graces but then they don't want to for example dismantle the architecture of using statistics so that's one angle which is like if you seek out the root of your descent with this kind of cognitive modeling you're going to be cutting off a lot of what is just understood to be statistical modeling so it's like you've gone so far into denying the map territory relationship that you've actually um biopsied more than just the fep and then they go further which is that um several features of the fep make it the ideal model of generic systems in statistical physics which is a bit of an abstract statement so we can bring it back to the textbook with figure 9.1 which just as um uh Michael brought up this came up in um the par recent bookstream with par where we we asked what was happening in this figure and and had some discussion um on it and here's the experimenter in the outer shell making a cognitive model that cognitive model of let's just say the rat and the teammates has an aboutness that reflects the experimental setting and context so it's making the experiments are human in this outer shell is making a

cognitive model which is a map so all the caveats of linear regressions apply maps not the territory don't want to fall into the map however it's also the case that that cognitive model that map is map like in its relationality to the experimental context so there's this view from the outside Black Box cognitive model being proposed as a hypothesis discrete Time chapter 7 continuous Time chapter eight whatever it may be but also the aboutness of that map is about this generative process which is the generative process handing stimuli to the generative model of the agent and so that duality view from the inside and view from the outside makes the FTP uniquely the ideal model of generic systems so they're saying yeah for sure don't do the naive fallacy but don't engage in the fallacy fallacy and then one final piece is if anyone has a dissent with this let's just see their alternative let's not simply tolerate a standalone criticism which is actually like a space for improvement but is anyone seriously proposing an alternative to this structure and the answer is not really Ali I think it's just to add a slightly more detail to uh with you what you just explained in terms of the paper itself well they actually lay out two lines of arguments against math territory fallacy one is as I mentioned uh the misconception that the process and the model or the map and the territory are conflated which they say they're not actually conflated but the territory behaves as if it were a map so it's a kind of nested representation and the other line of argument is as Daniel mentioned rises from a kind of misunderstanding of the uh the kind of modeling used in statistical physics and in particular the maximum the principle of Maximum entropy to model the environment so the way that uh I mean in fep the way that agent models or self evidences the environment is through cmap or a constrained maximum entropy principle so yeah I just wanted to add those two bits of detail there nice Michael so isn't isn't this that a map and a territory implies that there is an isometry between the map and the but in free energy um in certain and several points they claim these things are independent so the the model the process and the model is actually independent through these Markov blankets and so on and we cannot actually know what happens outside the world which is quite different from that idea of map and territory well great great questions I think there's two there's two threads there that are very tightly intertwined but they're really important to to consider separately so first you said is there an isomorphism or an isometry between the map and the territory isomorphism in the strict sense meaning identical structure is not required the generative process even just talking about the map the general process could be a continuous variable and the generative model could be like discrete or a binary categorical variable so isometry is not even required between the internal States and the external States within the map and then it's an even broader space where you

don't even know if isometry even applies to for example the structure of the real territory so for sure isometry or isomorphism is not required some kind of adaptive resemblance will come to be seen that's like the good regulator theorem from cybernetics any regulatory adaptive system is as if that epistemology that Ali mentioned it's as if it has an effective model of its Niche again that doesn't need to be isomorphic Evolution or far from equilibrium thermodynamics does make it the case that you're going to get swept off the table if you can't adapt and so over generations and through development it'll come to be the case that there's functional relationships and a semiotic process but it doesn't have to be as stringent as an isomorphism and the second aspect was the statistical Independence that is the technical definition of a Markov blanket so Markov blanket it's the nodes upon which two other sets of nodes in a Bayesian graph are conditionally independent so the fact that we're invoking the base graph right off the bat we know that we're talking about the map so territories don't have Markov blankets Maps do so already we're within the map discussion space and with the Markov blanket does it preclude Direct consequences direct causal impact from internal and external States so no telepathy no telekinesis it's just a model constraint however still through the blankets there can be the propagation of causal influences on a base graph for example a light could turn on in the generative process that could shine photons onto the set State and then that could update the beliefs of the internal States implications can still propagate on a base graph through the conditional Independence of the Markov blanket just like person A could tell person B and then person B could tell person C and that doesn't mean that person A told C yes Michael go for it are you so you're talking about the free energy principle but does it uh does this apply also to the notion of maps and territory as for instance we I mean Google Maps for instance they should have a more strict notion of isometry I guess uh I'm I'm it's just a question of what you were talking about about the free energy principle or the notion of maps and territory in general to what extent does this metaphor do you think that this metaphor really applies to arbitrary depths I I'm just for those who are watching this video just have to like my screen is like frozen so I'm just needing to kind of um reset the screen share but um uh I mean arguably Google Maps it depends on what you mean by an isometry that's a two-dimensional GPS landscape so some kind of distorted representation like making it flat is done within the bounds of what's useful I'm not sure what what else needs to be claimed for the purposes of Google Maps okay and and so you're saying the same also applies to this uh these Markov blankets and the free energy principle what do you think Ali again we go back again and again to purses taxonomy of symbols um actually

Maps so either I don't know Google Map or circuit diagrams or these kinds of map can act as a kind of icon for the object instead of the index so sorry I thought I got disconnected um icon instead of index so an icon is a sign that stands for the object um through the relationship okay so uh this kind of um relation between the icon and the object is a kind of structure preserving mapping especially in the case of geographical map maps or any other cartographical different mappings but uh in my opinion when we talk about maps and territories in relation with physical systems or biological systems or whatever we use this term in a slightly more uh broader and a much more looser term and it doesn't require to be an exactly structure preserving mapping so it's more like a mathematical mapping between a mathematical relation between a sets of object or sets of states and the other sets of States namely generator processes and the generative models so that's it I mean there isn't a who need any need to impose any additional requirement for uh the exact structure of those mappings it's a general mathematical mapping per se awesome yeah one other piece like from the empirical practice side that I think informs our philosophy of science is very rarely is the empirical researcher simply interested in producing one final model ing is almost always an iterative process that involves the maintenance and diversification and selection from portfolios of models so the findings of a paper might be like this phase space that we explored of models these structural families suggest that the two factors are better than the one factor or that short-term memory does interact with valence in this way so right off the bat empirically we're talking about portfolios of models hence there's obviously a multiple to one mapping of maps to System of Interest Ergo it's a bit Limited to try to over interpret any map and even get into this discussion of whether it's actually the territory it's not that's why we're doing it that's statistics that's modeling we have a whole host of models on our computer none of them are the territory but it's so tantalizing because first off unlike the linear regression we're talking about the action perception Loop which brings this sort of tangibility within our grasp um also there's this question of language with the instrumentalism and the realism being mixed together conceptually and linguistically well the agent is minimizing free energy is it though or is it the modeler's generative model of the actual embodied organism somebody who knows that may be able to speak or interpret flexibly and understand that when people are talking about an organism minimizing surprise or the imperative for an organism to minimize surprise be able to on one hand lean into the realism and on the other hand lean into the instrumentalism but neither of those I guess is like a standalone interpretation or or straightforward implication from any set of formalisms let's go to chapter eight and let's see how it plays out with

continuous time as it's our um first discussion in eight Let's uh scan through and then we'll come to the questions next week Okay so everything flows nothing stands still one of the cherished expressions of process philosophy and dynamicism chapter 8 complement 7 by continuing the discussion of generative models chapter 7 was about discrete time chapter 8 is on continuous State space models and it'll be brought up in multiple ways that continuous time models continuous perceptual models and continuous action models are often well suited for proximate generative models not using a term formally but just like dealing most proximally with incoming sensory data and on the action side dealing most proximally with motor actuation turns out that those settings often are continuous in nature like the amount of pressure on a proprioceptor or the amount of volume by an auditory sensor on the sensory sides and then on the action side like the angle of an elbow or the amount of force being exerted by a muscle those perceptual and motor settings are really amenable to continuous modeling whereas discrete time models are often amenable to categorical symbolic and discrete settings section 8.2 and just raise your hand if you have like a want to add something or a question 8.2 from chapter four the generative model in continuous time can be written as a pair of stochastic equations determining how States hidden States X generate data observables y so that's the top equation is the data are being generated by a generator function on x hidden States plus a noise term on y and change in x $\dot{x}$ derivative on X is a function of the current state of X and V which are slowly varying causes or contexts plus a noise term on x this is a classical formalization and in the context of SPM and brain Imaging this is like why is the fmri or the EEG measurements X are the unobserved neural States neural activity Dynamics and there are some sensor noise and then the Dynamics of neural activity which again are unobserved inferred from data are a function of current state of those neural activity patterns and slowly varying causes which might be like an attentional factor or might be an experimental factor and more broadly just this partition into like signal and Noise comes up again and again Ali I just wanted to point out the the second equation is also called Edo stochastic differential equation so if someone come across that term that's exactly this second equation of 8.1 awesome these stochastic calculi help us do integrals and derivatives on stochastic processes action is absent from equation 8.1 this is because as outlined in the recipe action is part of the generative process not the generative model so this is potentially like saying the consequences of action are part of how things are actually generated which is to say the generative process even though action selection is a function of a generative model the consequences of action in terms of modifying how the world changes are occurring in the

generative process if we write down the Dynamics of the real world the generative process but of course this is like the Dynamics of the map of the real world we would have to include action so we have a very similar equation but U is now replaced uh for V and there's bold G and F this is a general formalization with this sort of observable layer caused by hidden States and the change in the hidden States being related to their current state and causal factors they're going to specify in equation 8.3 a simple version that's compatible with the above and noiseless so it's a point attractor when X is less than V the expected rate of X is positive and vice versa so this is a set point attractor they're going to define a generative process where action will change to fulfill the predictions of 8.3 so this is an active movement towards a point attractor so that point attractor might reflect a preferred Target for the eyes or a preferred location in space for a limb this is going this is bringing us from this totally general setting into an active attainment or an active agency exertion that's compatible with those formalizations they're going to take it one level more complex a more sophisticated model recognizes that forces generated by muscles change velocity not the position so you don't just pick up and move your limb you're actually modifying a derivative of position which is Excel Excel velocity and acceleration and so here we start to see a little bit of what could be extended into higher and higher derivatives with the generalized coordinates of motion but now f of X and V which is to say the function that is a function of the current state of the Hidden state and the slowly causal uh varying factors are going to be explicitly uh written as this X_1 is the position and X_2 is the velocity now this generative model is describing Hook's law so this is a spring movement with with a conservative spring we can write down a general model that predicts the Dynamics that wouldn't fold if there was a spring drawing a limb to a desired location so that's like one level more advanced than the uh Point attractor which just linearly picks up the arm and moves it to the point now we have like this hooke's Law Spring Force drawing and oscillating and then we can imagine you can keep on going you can have a damping spring and so on 8.3 continuous time formalizations are well suited to characterization of movements and really just non-linear dynamical systems wherein discretization of time and space is efficient #livestream 52 discretization algorithms and optimization figure 8.1 we're returning to the spinal reflex case from chapter five descending predictions getting mashed up with incoming proprioceptive data resulting in an error residual that is modifying the trajectory of the um actions in the face space shown here X is the position of the limb and V is where it's being drawn to X is current position V is the Target location 8.1 Precision attention and sensory attenuation we could talk more about it but this

related to a discussion that we had in office hours last week about the uh sensory attenuation here are some of our discussions we talked about why it's relevant to model which is like the reduction of precision on proprioception in order to facilitate action selection and that's also kind of described here High precision sensor input you just get shut down no movement is executed it's catatonic they're going to move into two classical dynamical systems one with uh Predator prey like Dynamics lock cover Volterra and the second with chaotic Dynamics Volterra Predator prey model also called a winnerless competition equation 8.6 gives the lock of Volterra Dynamics and here on the top is the winnerless competition amongst plant herbivore and carnivore also one could imagine this is like activity patterns in the fmri OR EEG with different brain regions and that's exactly how it's been used and here is the phase space this is like three three lines plotted on time axis but we can also think about that as a trajectory in a three-dimensional space so some sort of Cube and these there's just one point a b c or phc that are moving in this three-dimensional space on some trajectory and then here if you look at one side or the other of a cube you can look at how there's something like a a slowly drifting limit cycle when you project down to a given axis now they're going to move that lockabletera winnerless competition Predator pre-dynamic into sequential action here's how sequential action has been used in iblink conditioning experiments and in writing generation of sequential information patterns and also in the par bookstream 2.1 there was a great discussion by Thomas so he said early octave it was continuous it was generalized filtering and then there was the engagement with action people started thinking about how to get sequential Dynamics emergence of sequences from continuous Dynamics was a key step towards purposeful planning that's figure 8.3 then it's like okay that's all good for making this kind of semi-regularized scribble but what if you wanted actual policy selection on which word to write that's going to be really hard to bake into a lock of ultra model things moved into the explicit discrete State space model with the pomdp formalization and that led to planning as inference and all these other features and now we're moving back into these hybrid and hierarchical models where continuous State spaces are reintroduced as a low level of hierarchical modeling 8.2 learning continuous models continuous State spaces are a lot like discrete State spaces in discrete State spaces you're basically taking a sum across a finite number of entities in continuous State spaces you're taking an integral from here to there and there are different approximations that facilitate those integrals the pmdp formalization of chapter 7 has largely superseded the use of generalized lockabletera systems in active but it's still extremely didactic and pedagogical and used in the lower levels of hierarchical models

now they're going to move from the lockableterra to the Lorenz attractor system which has some parameters that are reflective of weather systems that's been studied in the context of Birdsong and they're going to use the Lorenz attractor in the Birdsong case to talk about generalized synchrony so generalized synchrony doesn't mean lock step synchronization it means that there are entities whose Dynamics are on a manifold in other words their Dynamics are not untethered from each other there's Mutual information such that they're co-ambulating on a lower dimensional manifold so the Birdsong example has been in multiple papers it's a chaotic attractor that's underlying the emission of Birdsong by the two birds livestream 32 is on couples chaotic systems so that's where the math are unpacked the most but here we can just see on the right side of figure 8.5 that after learning there's a tightening of the first and the second Bird's expectations towards a manifold it's called singing from the same hymn sheet briston's quite fond of that phrasing and each bird contributes sections of the song and there's some interesting uh connections back to sensory attenuation again section 8.5 goes into hybrid discrete and continuous models which we've basically been talking about multiple times so figure 8.6 first off note that the G should be connected to the pi then note the fractal rake structure we have the rake here figure 4.3 on the bottom continuous time $v \times x \times \text{Prime double Prime}$ and so on but these continuous time models lower half of figure 4.3 are embedded within a discrete time higher level $a \ b \ d \ s \ t \ \text{minus } 1 \ t \ t \ \text{plus one}$ we see both parts of figure 4.3 discrete time on top $d \ b \ a$ past present and future and now we're seeing it as a top level nesting continuous time lower level model and here x derivative of x second derivative on x key difference between discrete time and continuous time models yeah I have a question what happens if the embedded time frames are not fitting into the upper frame so if for instance here the lower parts overlap with the um with the states of the upper part so if you have a time frame and maybe and chopping up of time into segments but the segments overlap is there a possibility to model such kinds of things yeah a few thoughts good question the first is that the continuous time model doesn't make explicit predictions about the future The Continuous time model is a lot like a Taylor series approximation so although it embodies through the generalized coordinates which are one two and three shown here but could be up to six or more it embodies some patterns that extend beyond the presence but the continuous time formalization is actually always real Time That's one answer a more empirical answer is if you have a model um there's like sort of clashing time scales you're going to end up with multicollinearity in fitting the data to empirical results so you'll get a variety of model fitting aberrations because the variance is going to be partitioned in a sort of uh discordant way so

some random seed might partition a lot of the variants up here and then another random seed partitions it down here so if there's Discord and overlap you're just going to end up with a headache trying to get that model to be useful so no one's going to stop you but that doesn't mean that it's going to be an effective architecture for actual parameterization so does it mean we need to squeeze somehow the stuff into these boxes into these rectangles embedded rectangles there's no not really a way to have these overlapping boxes there what's the case that we're thinking about because pretty much here is the movement of time at some click and then this could be every second is when the new limb location is chosen and then within each second there's the continuous movement of the limb yeah so so for instance they talk about a sentence so we have words and sentences paragraphs um texts and so on but now if you look at prosodic features they may overlap with sentence boundaries so you wouldn't have a kind of pause structure that fits into the sentence structure there could be two different ways of chopping up the sentence in an acoustic way and um well in a you know in terms of words and sentences and so on I think it's a super interesting question which is like this strict nesting like well Isaacades continuously let's just say and they're recognizing letters okay letters are strictly nested within words words are strictly nested within sentences paragraphs and so on and one might want to go oh and paragraphs are strictly nested within narratives it's like well no not necessarily there might be narrative boundaries that are up down left right of the paragraphs so let's see it start by just graphing it how you see it and then let's go from there but in principle it's not going to be addressable but when you actually write it out you may discover again a portfolio of model structures which can be used empirically to differentiate hypotheses and their Maps so maybe it turns out that a simpler model where prosody is sentence delimited is more effective because there's so fewer parameters so the Bayesian information Criterion favors that model but that doesn't mean that prosody is sentence delimited it means that a statistical structure with sentence delimited prosody was selected for information theoretic purposes map territory but we want to say things about the so that's always the challenge we do a regression between foraging activity and the humidity and we want to say when it's more humid the ants forage more that seems like the natural thing to say and it might be our motivating question but it's always hard to remember that it's like ah higher values in the humidity measure are correlated with higher foraging activity it's like just say that foraging increases with more humidity but that is the departure point of the map territory it's often more resolvable than people want to know in the specifics that's the deflationary perspective and less resolvable than we want to know in principle Ollie and also one reason

behind formulating the continuous Dynamics in terms of a Taylor series is because it also Maps uh onto the lagrangian formulation of mechanics because one of the reasons lagrangian's mechanics is used instead of the traditional Newtonian mechanics is to just get rid of time parameters so instead of time they use momentum and also the lagrangian so we don't need time explicitly defined or described in order to account for the behavior of the trajectories or to describe the behaviors of the trajectories what all we need is this series of successive differentiations in terms of Taylor series to be able to describe the whole behavior of the system because accounting explicitly for time parameter is much difficult to model than than describing these dynamical systems in terms of their generalized coordinates in other words they're lagrangian formulations and in the path integral formulation of fep that's basically the solution proposed by Dalton person and others to use lagrangian formulation to describe these systems nice good point equation 8.8 is describing the transformation from the discrete Target selection of the upper level of this model and the way that that is received by The Continuous time model so here the discrete time Target selection is then received by The Continuous time icicade model that is an experimental setting that has been studied relatively extensively here in the middle of figure 8.7 we see the target selection being modeled and the cicading and here's the calling back to that spring-like movement here's like the acceleration and then the dampening as the target is fixed upon so isocating has been studied relatively extensively in the context of information driven box 8.3 is a bit of a side note on discretization so a gaussian mixture model or very related is a k means model where one takes a cluster of data points in space and asks how many clusters k fit scattering of points you could just have one K with just a point in the center and then there's some variance distribution around it or you could say 2K and then you can look at the error profiles to determine uh a adequate number of clusters somewhere between one cluster and one cluster per data point there's going to be a k means value that is like statistically optimal 8.6 in this chapter which is relatively short and in some ways kind of introducing the continuous time formalism so that we can get to these important topics like Hybrid models they overview applications of continuous time GMS this is a huge topic and much has been left out see table 8.1 synthetic bird song ocular motor delay conditioned reflex smooth Pursuit eye movement 2012. psychosis illusions 2012. cicades many papers action observation attention hybrid discrete continuous models self-organization it's a foundation for further exploration they focused on movement prediction it simplifies the treatment of motor control we don't need any additional Machinery or inverse models and this was also a topic picked up in the skilled performance discussion of hippolito

at all talking about interactionism rather than instructionalism they talked about the lock of Volterra winnerless competition Predator prey and the lorenza tractor chaotic system in all those cases it was possible to find this concept of generalized synchrony which isn't lockstep but it's like a mutual information or uh manifold controversial and then they introduced the Hybrid models which tend to have continuous lower level Dynamics and discrete higher level Dynamics that's it for today next time we're going to come back to chapter nine which is going to um be talking about model based data analysis fitting models to data next week uh we'll do chapter 8 questions and come through it again in preview nine so thank you fellows see you next time